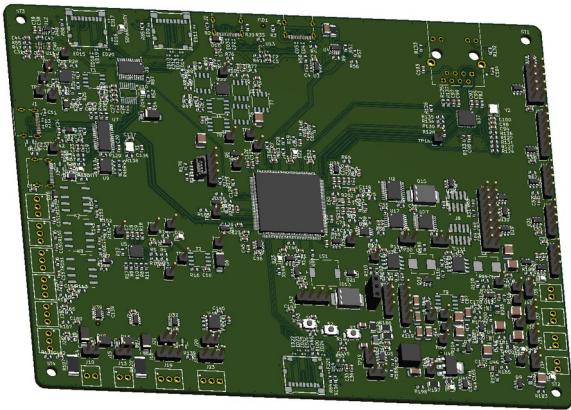


PhysIO

Gives AI coding agents hands on the embedded bench

RUFILLA

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OVERVIEW

Hands-free flash, boot and debug

AI agents write firmware, then stop at the board: every flash, power cycle and log capture needs a person. PhysIO puts the target under the agent's control over one USB-C cable.

Plain-text SCPI over USB serial, no drivers. An MCP server exposes it to Claude and other agents as tools.

COMPATIBLE WITH



KEY FEATURES

- **Flash without hands** Write an SD image at USB High-Speed and boot the target.
- **Every bus, every voltage** UART, SPI, I²C, GPIO, CAN-FD, RS-485 at 1.6–5.5 V.
- **Measure and stimulate** ADC inputs, DAC waveforms and three relays.
- **Power under control** Switch and current-limit four rails; log current draw.
- **Built-in debug probe** SWD for OpenOCD, probe-rs and pyOCD, plus a Cortex header.
- **labgrid compatible** Power, reset, console and SWD for your CI rig.

OPEN SOURCE HARDWARE & FIRMWARE



Open hardware

Schematics, PCB layout and BOM under CERN-OHL-W-2.0



Open firmware

Zephyr and Rust firmware under Apache-2.0 or MIT



Open host tools

MCP server and labgrid drivers under Apache-2.0 or MIT

SPECIFICATIONS

Technical specifications

CONTROLLER

Processor	STM32H723, Arm Cortex-M7 at 550 MHz
Firmware	Zephyr RTOS, with Rust application code
Host link	USB-C: USB serial (CDC) and CMSIS-DAP; optional 10/100 Ethernet

POWER

Input	USB Power Delivery from the host cable or a dedicated PD input, selected automatically
Target rails	1.8 V; 3.3 V at 2.48 A; 5 V at 2.04 A; 9 V at 1.37 A (needs a 12 V PD source). Each rail switched and eFuse-protected
Monitoring	3× INA219: USB passthrough and two user-wired channels

TARGET INTERFACES

Buses	2× UART, SPI, I ² C and 8-bit GPIO; CAN-FD to 5 Mb/s; RS-485 (Modbus RTU)
I/O levels	1.6 to 5.5 V, referenced to the target's VREF; tri-stated until it is valid
Debug	Integrated SWD probe; 10-pin Cortex Debug header
USB	USB 2.0 High-Speed passthrough with switched, current-limited VBUS

STORAGE & I/O

microSD	Shared card with USB High-Speed reader and two-way mux to the target; separate card for soak logs
Analogue	16-bit ADC inputs; 2× 12-bit DAC with sine, ramp and square generation
Bench	3 relays, OLED status display, three buttons, buzzer, Qwiic expansion

HOST SOFTWARE

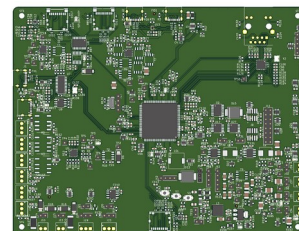
Protocol	SCPI text commands over USB serial; no drivers
Integrations	physio-mcp (MCP server for AI agents); physio-labgrid (labgrid resources and drivers)

STATUS & COMPLIANCE

Development	Firmware and MCP server bench-validated on STM32H723; v1 PCB in layout
Use	Development tool for low-voltage targets only; not CE, UKCA or FCC certified

MORE ONLINE

Project page, updates and the latest version of this datasheet:
labs.rufilla.com/physio



Talk to us about
PhysIO



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Get started

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